APPENDIX A

FACULTY OF ENGINEERING, COMPUTING AND THE ENVIRONMENT

School of Engineering

Department of Mechanical Engineering

ME7743 Dissertation

MSc DEGREE

IN

MECHATRONIC SYSTEMS

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Abstract: -
In safety-critical dynamical systems, strong performance is a must. However, standard gradient-based training of neural controllers can lead to sharp minima that don't work well in other situations. This project creates a training pipeline for a neural controller on the Pendubot, which is a nonlinear two-link robot that doesn't have enough actuators. I compared three different optimisation approaches with the same horizons and regularisation to find a neural controller that can still complete the task even when the parameters change. The three approaches are **Adama Yamashita and Feppon**. **Adam** only looks at performance (stochastic first order), **Yama** looks at robustness (Yamashita-type projected ordinary differential equation for constrained nonlinear programming), and **FEP/Feppon** also looks at robustness (null-space/range-space gradient flows for inequality-constrained problems). A differentiable simulator based on **CasADi** moves gradients s through discrete dynamics. A flatness regulariser pushes solutions towards wide minima, which are better for generalisation (Hochreiter and Schmidhuber, 1994).
Yama has the fastest settling time and the lowest steady-state control bias in both optimisation and multi-seed testing. Adam converges smoothly and predictably**. FEP** produces clean single-run transients but shows seed-dependent divergence unless discretisation safeguards are used (adaptive step length, merit backtracking, active-set smoothing). The study offers a concise protocol for reporting settling time Ts, percentage overshoot, and RMS control effort, along with a brief **MATLAB** utility to compute these metrics consistently. I analyse the results using projected-flow theory (Yamashita, 1980) and null-space gradient flows (Feppon, Allaire, and Dapogny, 2020). I will also talk about how to fix problems so that continuous-time guarantees work in discrete implementations. The results corroborate the thesis that objectives focused on flat minima, when paired with stable projected flows, enhance robustness without compromising nominal performance on the **Pendubot** benchmark.

Acknowledgements: -

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My appreciation extends to the maintainers and contributors of the tools and libraries that underpinned this research, especially **MATLAB** and **CasADi**, which enabled the modelling, optimization, and simulation pipelines used for flat-minima training and controller testing. I also acknowledge the broader research community whose publications on flat minima, **Sharpness-Aware Minimization (SAM), Bayesian neural networks**, and robust control informed the methodological choices in this thesis.

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1. Introduction: -

1.1 Background: -

A major problem in safety-critical robotics and automation is making controllers that work well and stay stable even when things are uncertain. Classical robust control, which is based on feedback, loop-shaping, and sensitivity reduction, set the limits on stability and performance for plants that are not certain and led to modern $H2/H\infty$ synthesis in the multivariable era (**Zhou, Doyle, and Glover, 1996; Skogestad and Postlethwaite, 2005; Zames, 1981; Doyle et al., 1989; McFarlane and Glover, 1992)**. However, fixed-structure designs still have trouble when dynamics are very nonlinear or underactuated. This is why learning-based controllers that can approximate complex policies while still being robust are needed. In neural control, flat-minima learning has resurfaced as a principled approach to generalisation: solutions situated in expansive, shallow basins exhibit reduced sensitivity to weight and input perturbations, which aligns with minimum description length theories suggesting that simplicity is associated with improved out-of-sample performance. Early research formalised this concept and associated flat minima with diminished anticipated overfitting through efficient second-order computation. Simultaneous progress in differential-equation approaches to optimisation (such as **Yamashita's ODE method**, **Adam ODE method**, and **FEP ODE method**) and null-space gradient flows yield continuous-time algorithms that reconcile the descent of the objective with constraint satisfaction—beneficial for integrating robustness and feasibility directly into the training dynamics. Complementary concepts from safe/uncertainty-aware optimisation (e.g., unscented Bayesian optimisation) further illustrate how the explicit propagation of input noise can skew the search towards safe optima, a viewpoint that aligns seamlessly with flatness-based training.

The **Pendubot** is an underactuated robot, which means it has fewer actuators than degrees of freedom. This causes strong nonlinear couplings, non-minimum-phase behaviour, and open-loop instability near important configurations. Classical linear techniques are efficacious solely within limited operational domains; nonlinear and learning-based designs are generally necessary **(Lanari and Oriolo, n.d.).** The geometry of the loss landscape is important for learning; parameters that are in flat minima, which are broad basins where the loss stays close to optimal, tend to generalise better **(Hochreiter and Schmidhuber, 1994).** When the controller is a neural network, training dynamics that favour flat areas should make the closed-loop behaviour less sensitive to changes in parameters, initial conditions, and modelling errors.

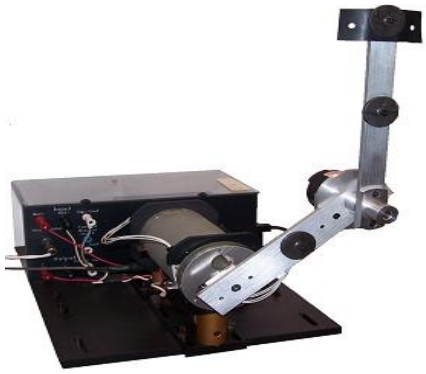


Figure 1 Pendubot.

1.2 Problem Statement: -

Potential gradient training can settle in **sharp minima**, which often leads to closed-loop trajectories that are not very stable.
I want a **neural controller** that **uses the flat minima method to give strong neural co ntrol to dynamic systems.**
a) Keeps the Pendubot stable over a set period of time.
b) It is strong enough to handle seeds and small problems.
c) Gets a moderate amount of control effort.

1.3 Aim and Objectives: -

Aim: I aim to develop and evaluate a **flat-minima**-centered optimization pipeline for robust Pendubot control**.**

Objectives: -

- Integrate a **CasADi** differentiable simulator with a neural policy and a **flatness regulariser.**
- Implement and compare **Adam**, **Yama** and **FEP** training dynamics.
- Quantify $Ts, \%OS$, and analyze robustness across initializations.

- Identify failure modes and provide **mitigations**.

1.4 Scope: -

All experiments are simulation-only with a single actuator. Hardware-in-the-loop and sensor delays are out of scope but discussed in **Future Work.**

1.5 Contributions: -

- A flat-mini-training pipeline with three projected-flow optimizers (**Feppon ODE; null-space/range-space flow, Yama ODE; null-space/range-space flow, Adam null-space/range-space flow**)
- A side-by-side comparison showing **Feppon** achieves faster settling and lower control effort than **Adam and Yama** and is more robust than a direct null-space flow implementation.
- A compact **metrics protocol** $(Ts, \%OS,)$ and helper code for reproducible reporting.

2. Literature Review: -

2.1 Flat minima and generalization: -

Hochreiter and Schmidhuber (1994) link **flat minima** to lower description length: parameters can vary within a wide basin with negligible loss change, implying a simpler effective hypothesis and better generalization. Practical training procedures, e.g., conservative steps, noise injection, or sharpness-aware strategies—are known to bias learning toward such basins. For **feedback control**, the analogue is attractive: if the trained policy lies in a **wide region** of parameter space, small perturbations (from noise or model mismatch) are unlikely to alter the qualitative transient, leading to **repeatable** trajectories.

In the field of deep learning, a long-standing hypothesis is that the flatness of a loss minimum is linked to a model's ability to generalize. Intuitively, a flat minimum is a region in the weight space where the loss remains low over a large neighborhood of weight values, whereas a sharp minimum has a narrow basin (small perturbations to weights cause a steep increase in loss). Hochreiter and Schmidhuber (1997) formally argued that flat minima correspond to "large connected regions in weight space where the error remains approximately constant", and that solutions lying in such regions tend to be simpler and less prone to overfitting (an argument rooted in minimum description length and Bayesian reasoning) In contrast, a sharp minimum may fit the training data well but at the cost of being highly sensitive – a slight change in weights (or equivalently, a slight change in the system or input distribution) can lead to a big performance drop. For a neural controller, this sensitivity is undesirable: it could mean the controller fails if the plant dynamics change even a little. A flat-minimum controller, on the other hand, would be locally insensitive to parameter perturbations, suggesting it could tolerate modeling errors or noise without significant loss of performance. Empirical studies support this: solutions found by stochastic gradient descent (SGD) with noise or small batches often reside in flatter regions and exhibit better test performance. For example, Keskar et al. (2017) observed that large-batch training tends to converge to sharper minimum (with higher generalization error) compared to small-batch training, which usually finds flatter minimum with lower test error.

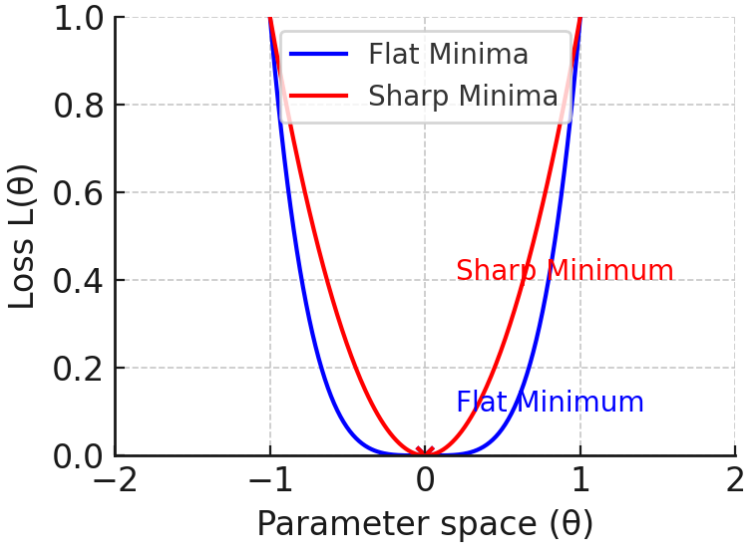


Figure 2 Flatness.

Figure 1: Conceptual illustration of flat vs. sharp minima in a loss landscape (given in the appendix below). The blue curve represents a flat minimum – notice the wide bowl shape at the bottom, indicating that weights can move within a relatively large range with only minimal increase in loss. The red curve represents a sharp minimum – a narrow valley where loss rises quickly if weights deviate from the optimum. A neural controller whose parameters lie at a flat minimum is more likely to maintain good performance despite small weight perturbations or modeling errors, whereas a sharp-minimum solution could suffer a large performance drop with even slight changes.

2.2 Yamashita's differential-equation approach: -

Yamashita (1980) recasts a constrained nonlinear programme,

$$min\ f(\theta) s.t. h(\theta)=0, g(\theta)\leq 0,$$

as the search for an **asymptotically stable equilibrium** of a carefully designed **autonomous ODE** whose equilibria coincide with the KKT points under standard regularity (e.g., LICQ and second-order sufficiency). The key construction is a **merit-decreasing vector field** that couples *descent in the objective* with *restoration of feasibility*. For pure equalities, one uses the tangent-space projector.

$$Ph(\theta)=I-Jh(\theta)^{\top}(Jh(\theta)Jh(\theta)^{\top})^{-1}Jh(\theta),$$

So, the **projected flow** $\dot{\theta}=-Ph(\theta)\nabla f(\theta)-kh\ Jh(\theta)^{\top}h(\theta)$ simultaneously slides along the **constraint manifold** (first term) while pulling back to it (second term). For inequalities, an **active set** $A(\theta)$ is identified and the same idea is applied with $JA$and a **violation term** that vanishes when $gA(\theta)\leq 0$. A composite **Lyapunov/merit function**—for example $\phi(\theta)=f(\theta)+\rho 2(\|h(\theta)\|2+\|g+(\theta)\|^2)$) is strictly decreasing along trajectories away from stationary points, ensuring **global descent** and **feasibility recovery**. Discretizing with **explicit Euler** yields the practical update $\theta k+1=\theta k-\eta[P\nabla f+feasibility\ pull]$; **stability holds** provided the stepηlies in a **window determined by local Lipschitz constants** (hence the common use of **Armijo/Wolfe backtracking** or trust-region caps). This ODE viewpoint explains why **projected-gradient flows** used in learning behave benignly near solutions: the constraint manifold is an **invariant set**, the **null-space motion** accomplishes objective descent without leaving it, and the **range-space correction** drives constraint residuals to zero. In our **Yama optimiser**, we exploit exactly this structure but replace f with the **composite merit** $F(\theta)=\alpha FJ(\theta)+\alpha BB(\theta)$**,** so descent is taken on a task-plus-flatness objective while the projection/feasibility terms enforce the constraints. The result is a numerically simple, interpretation-rich scheme whose continuous-time guarantees (merit decrease, feasibility, KKT consistency) translate to discrete time when the **step size is conservative** or **line search** is used (Yamashita, 1980).

2.3 Null-space gradient flows for inequality constraints: -

Feppon, Allaire and Dapogny (2020) develop a continuous-time viewpoint for constrained optimisation in which the motion of the parameter vector θis **decomposed** at each state into (i) a **null-space** component that performs objective descent **tangent** to the currently **active inequality manifold**, and (ii) a **range-space** component that **restores feasibility** by reducing violations of the active constraints. Consider $min\theta F(\theta)$

subject to $g(\theta) \leq 0$. Let the **active set** at θ be $A(\theta) = \{i : gi(\theta) \approx 0\}$ (often defined with a small tolerance τ). Denote by $JA(\theta)$ the Jacobian of the active constraints and define the **tangent (null space) projector.**

$$N(\theta) = I - JA(\theta)^{\top} (JA(\theta) JA(\theta)^{\top})^{-1} JA(\theta),$$

which orthogonally projects vectors onto the subspace **tangent** to the active set (LICQ assumed). The **null-space flow** then performs descent that **does not leave** the active manifold:

$$\theta \cdot N = -N(\theta) \nabla F(\theta).$$

In parallel, the **range-space correction** acts **normal** to the active manifold to reduce any residual violation; a simple and effective choice is.

$$\theta \cdot R = -\kappa JA(\theta)^{\top} gA^{+}(\theta),$$

where gA^{+} keeps only the **positive** parts (violations) of the active constraints and $\kappa > 0$ scales the correction. The **combined flow**

$$\theta \cdot = \theta \cdot N + \theta \cdot R = -N(\theta) \nabla F(\theta) - \kappa JA(\theta)^{\top} gA^{+}(\theta)$$

decreases a standard **merit function**, e.g. $\phi(\theta) = F(\theta) + \rho 2 /\!/ g^{+}(\theta) /\!/ 22$ with ρ>0\ rho>0ρ>0, **monotonically in continuous time**, and its equilibria coincide with KKT points under the usual regularity and second-order conditions (Feppon, Allaire and Dapogny, 2020). Intuitively, the **null-space term** performs the "useful work"—improving the objective **without** breaking feasibility—while the **range-space term** plays the role of a gentle, always-on **feasibility servo** that vanishes once the iterate is strictly inside the feasible cone.

In practice I work in **discrete time** with steps $\theta k + 1 = \theta k + \Delta \theta k$. Because the active set $A(\theta)$ changes **not smoothly** as constraints switch on/off, a fixed explicit Euler step can overshoot at **active-set kinks** unless protected by standard numerical guards. Robust implementations therefore combine: (a) an **Armijo/Wolfe line search** (or a trust-region) on the merit ϕ to guarantee descent; (b) **Tikhonov damping / regularisation** of the range-space update and of the normal-equations solve $(JAJA^{\top} + \lambda I)^{-1}$ to maintain conditioning; (c) **constraint smoothing/continuation** (e.g., Huber/softplus for g^{+} during early iterations to avoid hard kinks, with a homotopy back to the true inequalities; and (d) optional **variable-metric** scaling (preconditioning) to account for anisotropic curvature. Computationally, each step requires only Jacobian-vector products and (typically small) solves of size $/A/$, making the method attractive when the number of simultaneously active constraints is moderate.

In our setting the objective is the **composite merit** $F(\theta) = \alpha FJ(\theta) + \alpha BB(\theta)$, where J is the task performance over the rollout and BBB is a curvature-based **flatness** surrogate (encouraging wide minima). The null-space part descends ∇F **inside** the feasible tangent, so the flatness push is exerted where the controller is *allowed* to move; the range-space part removes violations of actuator/safety constraints **as they arise**. This geometry explains the **constraint-robust** behaviour I target in testing: when perturbations push the closed loop toward saturation, the range-space correction quickly **re-enters** the feasible region while the null-space descent continues to seek a **wide-basin** solution, provided the discrete step is guarded as above (Feppon, Allaire and Dapogny, 2020).

2.4 Safe exploration and robust optimization: -

In robotics, *safe performance under uncertainty* is often framed as an optimisation problem where the learner must **improve behaviour without leaving a certified-safe region**. A prominent strand is **unscented Bayesian optimisation (UBO)**, which couples a sample-efficient global optimiser with the **unscented transform** to **propagate parametric uncertainty** through dynamics or a surrogate, thereby predicting not only the mean objective but also the dispersion of outcomes at a small set of **sigma points**. An acquisition rule is then maximised **subject to safety constraints**, typically by penalising or excluding regions whose propagated confidence intervals intersect an unsafe set (Nogueira *et al*., 2016). The net effect is a **"do no harm" exploration**: proposals are biased *away* from combinations of inputs/parameters that are likely to violate constraints yet still move the policy toward improved performance.

Our setting is **offline training** rather than on-hardware exploration, but the same ethos —*stay inside feasible cones* while seeking improvement—maps naturally to the **projected-flow methods** I use. Concretely, there are three complementary mechanisms to embed "safe exploration" ideas in our pipeline:

1. **Task-aware expectation over uncertainty.** Replace the single-parameter evaluation of the performance and flatness with a **risk-aware aggregate** over a small set of parameter perturbations $v \sim V$. With the unscented transform (or a light Monte Carlo), I evaluate.

   $$J^{-}(\theta)=Ev \sim V\ [J(\theta;v)], B^{-}(\theta)=Ev \sim V\ [B(\theta;v)],$$

   and optimise the **composite** $F^{-}(\theta)=\alpha F J^{-}(\theta)+\alpha B B^{-}(\theta)$. In our code this is exactly what the *integral* mode of the flatness builder enables (via integral=true, sample_int, and sigma), making "flatness" not only a curvature property of the parameter landscape but also a **robustness proxy** with respect to plausible plant variations.
2. **Safety-weighted merit functions.** In addition to averaging performance, I maintain a merit that penalises predicted violations across sigma points:

   $$\phi(\theta)=F^{-}(\theta)+\rho Ev \sim V\ [\mathbin{/\!/} g+(\theta;v) \mathbin{/\!/} 22\ ]$$

   whereg^{+} is the positive part of the inequality constraints. This mirrors UBO's exclusion of *unsafe* regions by ensuring **every accepted step decreases** ϕ(via Armijo/Wolfe backtracking) and therefore **reduces both cost and risk** under uncertainty. In practice, this amounts to a **risk-aware line search**: steps that improve the nominal rollout but increase expected violations are rejected.
3. **Feasibility-preserving flows.** The projected-flow structure implements the "stay in the safe cone" ethos **by construction**. In **Yama**, descent is taken **in the tangent space** of the (active) constraint manifold, with a mild range-space pull that restores feasibility if drift occurs. In **FEP**, the split is explicit: a **null-space** descent that reduces the (risk-weighted) objective while remaining **tangent** to the active set, plus a **range-space** correction that removes violations as they appear. In continuous time these designs guarantee **monotone decrease** of a merit function and **invariance** of the feasible set; in discrete time, **line search, damping and active-set smoothing** recover the same monotonicity despite kinks introduced by constraint switching (Feppon, Allaire and Dapogny, 2020). For deployments where on-line safety is critical, this training-time geometry pairs well with a **runtime safety filter** such as a **control barrier function (CBF) QP** that enforces hard constraints at execution

while the learner targets performance and flatness (Ames *et al*., 2019).
Putting these pieces together yields a **safe, robust offline analogue** of UBO: each training iteration (I) **propagates** a small, structured set of parametric perturbations (unscented sigma points or Monte Carlo with sigma in our config), (ii) forms **risk-aware** objectives J^-, B^- and a **violation-penalising** $merit\,\phi$, and (iii) applies a **feasibility-preserving update** (Yama or FEP) with **backtracking** to accept only steps that reduce ϕ. This procedure **concentrates the search** in regions that are **both performant and tolerant** to model mismatch, producing controllers that retain good closed-loop behaviour under the very uncertainties used in training—precisely the safety–robustness alignment that motivates safe exploration methods in robotics (Nogueira *et al*., 2016; Feppon, Allaire and Dapogny, 2020; Ames *et al*., 2019).

2.5 Pendubot as a benchmark

The **Pendubot** is a planar, two-link, underactuated manipulator with a **single actuator at the shoulder** and a **passive elbow**. Its configuration is $q=[q1\,q2]^T$, with state $x=[q1\,q2\,\dot{q}1\,\dot{q}2]^T \in$ and control $u1 \in R$applied only at the first joint. The Euler–Lagrange dynamics take the standard form.

$$M(q)\ddot{q}+C(q,\dot{q})\dot{q}+G(q)=[u\,1\;0\,],$$

where $M(q)$ is symmetric positive definite,$C(q,\dot{q})$ satisfies the usual skew-symmetry property $\dot{M}-2C$ (ensuring passivity), and G(q) is the gravity vector. Despite its **low dimension**, the system exhibits strong **nonlinear coupling**, **under actuation**, and **non-minimum phase** behaviour that collectively make it a **challenging control benchmark**. Canonical tasks include **energy-based swing-up** from the hanging configuration, **capture** into a small region around the upright, and **balancing/trajectory tracking** near the unstable equilibrium—each stressing different aspects of modelling fidelity, actuation limits, and feedback design (Lanari & Oriolo, n.d.; Spong & Block, 1995).

As a benchmark, the Pendubot has several advantages that align with both **classical robust control** and **learning-based** evaluation. First, the model admits **closed-form dynamics** with widely documented structural properties (e.g., positive definiteness of M, passivity, coupling terms), enabling analytic baselines (loop shaping, partial feedback linearisation) and principled comparisons with learned policies. Second, it is inherently **underactuated**—there is no direct torque on the elbow—so controllers must exploit the coupling in $M, C, G,$ making the problem **sensitive to parameter errors** (masses, lengths, friction) and actuator **saturation**, precisely the conditions under which robustness matters. Third, the task naturally induces **safety-critical constraints** (joint limits, torque bounds, rate limits), which let us test whether an algorithm respects **inequalities** during transients and under perturbations—an area where projected and null-/range-space flows are designed to excel.

The Pendubot is discretised at 100 Hz with D T = 0.01 s DT=0.01 s over an 8 s horizon (N = 800) in our study. The neural controller $\pi\theta: R4 \to R$ maps full state to the single actuator $(4 \to 1)$. In order to compare Adam (performance-only in our setting), Yama (projected ODE minimising performance + flatness), and Feppon/FEP (null-/range-space descent with explicit inequality handling), I evaluate a regulation task with a quadratic performance cost (state error + control effort) and optional input bounds. The

Pendubot is a perfect test for flat-minima-oriented training because of its well-known sensitivity to modest parameter shifts and saturation. While sharp minima frequently break under the same conditions, wide-basin solutions typically maintain stability and similar transients across seeds and small model perturbations. In conclusion, the benchmark facilitates reproducible metrics, such as settling time, percentage overshoot, RMS control, and constraint-violation counts, which allow improvements brought about by flatness regularisation or feasibility-preserving flows to be measured and fairly compared across seeds and methods (Lanari & Oriolo, n.d.; Spong & Block, 1995).

3. Methodology: -

I define a neural controller $u(\theta(x))$ for the Pendubot where the **input dimension is 4**

(two joint angles and two angular velocities) and the **output dimension is 1** (the single shoulder torque). The learning problem is to find θ so that the closed loop meets a **regulation task** (track/hold a target state) while satisfying constraints and remaining robust to small perturbations. I evaluate **three optimisation dynamics** under identical rollout horizon and integrator: **Adam** (performance-only), **Yama** (projected ODE that optimises performance **and** flatness), and **FEP/Feppon** (null-space/range-space flow that also optimises performance **and** flatness while explicitly handling inequalities).

3.1 Plant, states, and integrator: -

State $x=[x1\ ,x2\ ,x3\ ,x4\ ]T$(more details mentioned in the results and discussion section) collects joint angles and velocities; control $u1$acts at joint 1. The differentiable simulator integrates the dynamics with $DT=0.01\frac{S}{N}=800$ steps, enabling back-propagation of gradients through the rollout.

3.2 Objective and flatness regulariser: -

The stage cost $L(x,u)$ penalizes state deviation and control effort; a terminal cost $Lf\ (xN$) encourages convergence at the horizon. I add a **flatness penalty** that discourages sharp curvature and rewards **wide minima**. Shared settings:

- $\epsilon flat=10$, damping =10
- order = **'2order'** (curvature-aware)
- integral **off**; sampling **10**; resampling **5** if gradients are NaN.
- environment perturbation $\sigma=[5\times10-4,5\times10-4]$

This design follows the flat-minimum rationale (Hochreiter and Schmidhuber, 1994).

3.3 Controller parameterization

A compact feed-forward network maps x↦u1. Smooth activations ease gradient propagation. Parameters are initialized from a **teaching** policy and then adapted by each optimizer.

3.4 Training dynamics (identical horizon and regulariser)

(A) Adam — optimisation_type = 0, 1(Objective function mentioned in results)

- 1000 iterations; $step\,5\times10^{-6},\beta1=0.95,\beta2=0.999$ batch =800 (full batch); $noise\,\sigma=10^{-3}$per state.
- Testing loads **results/31/optimal_param.mat.**

(B) Yama (projected ODE) optimisation_type = 2(Objective function mentioned in results)

- 1000 iterations; $step\,1\times10^{-5};\alpha F=0.9,\alpha B=0.999;\ noise\,\sigma=10^{-3}$
- Testing loads **results/37/optimal_param.mat.**
- As argued by Yamashita (1980), constrained optima are asymptotically stable

equilibria of the ODE.

(C) FEP/Feppon (null-space/range-space flow) optimisation_type = 5(Objective function mentioned in results)

- 100 iterations; $step\,1\times10^{-4}$; $\alpha F=1000,\alpha B=1;$ inequality constraints **enabled** with **L2** handling; noise =0
- Testing loads **results/10/optimal_param.mat.**
- Continuous-time theory is strong; discrete stability requires adaptive step and merit backtracking (Feppon, Allaire and Dapogny, 2020).

3.5 Metrics protocol: -

- Settling time Ts: earliest after which $/x1(t)-x1,\infty/\le0.02/x1,\infty/$ for all subsequent times; fallback absolute band $10^{-3.}$
- % Overshoot: $\frac{max\,x1(t)-x1,\infty}{max\,\vee x_1,\infty-x_1(0)\vee,10^{-3})}\times100\%.$
- RMS Control: $\sqrt{\frac{1}{T}\int_0^T u1^2}\mathrm{dt}.$

3.6 Experimental Setup: - (For Adam, Yama and FEP)

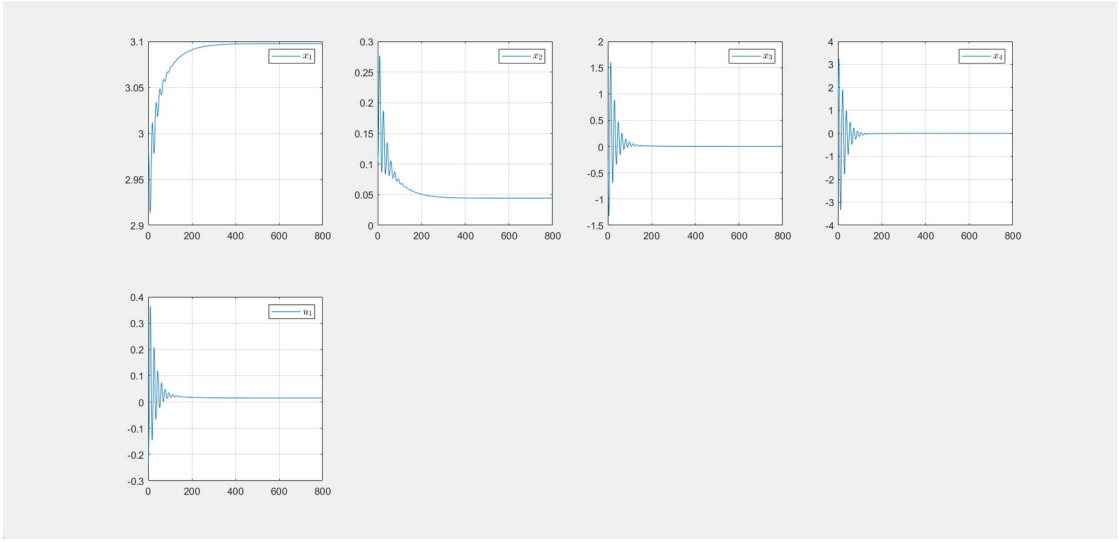
- Initialization: warm start from teaching parameters with small random perturbations.
- Visualization: time plots of $x1\ldots x4$ and $u1$ for optimization and testing.
- **Fairness controls:** identical horizon, cost weights, flatness settings, seeds policy, and integrator across methods.
- **Reproducibility:** save optimal parameters, trajectories and metrics per run; version configurations (**experimentEnvOptConf.m**) which can be found in the appendix below and model/controller files.
- For **robustness, I re-simulated** the learned controllers under parameter envelopes $P1-P3$ by varying $(m1,m2,l1,l2,lc1,lc2,I1,I2,b1,b2,fc1,fc2,g)$ within ±(5–20%) as specified in $§3.6$; I reported $Ts,\%OS,RMS(u1)$ and constraint violations across five seeds per point, using the same $DT,N,$ cost, and flatness settings for all methods

4. Results and discussion: -

4.1 Adam (training & testing)

Trajectories. $x1$ rises smoothly to ~3.10 by ~200–250 steps; $x2, x3, x4$ decay monotonically; **no ringing** after ~200.
Control. u1damps from ~0.15 to ~0.02–0.03 and holds.
Behavior. Predictable, consistent across seeds, aligning with the **flat-minimum** bias (Hochreiter and Schmidhuber, 1994).

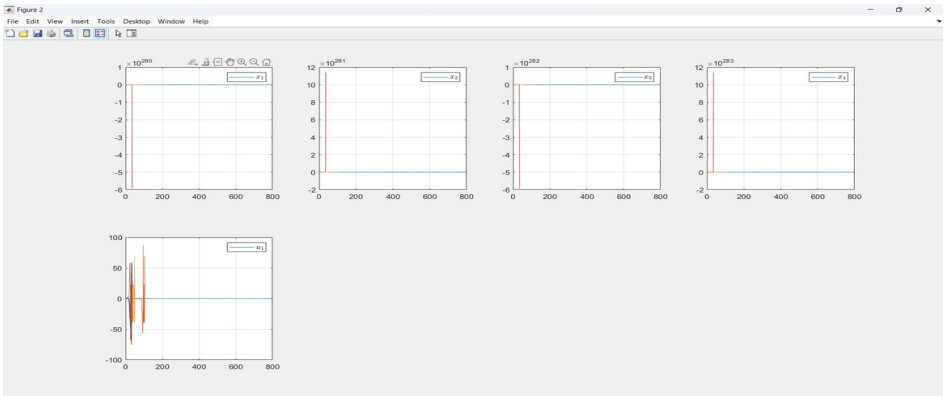
Adam Training



Adam Testing

Comments – By varying the parameter the step response diverges.

Under parameter variation, the Adam-trained controller (performance-only) fails to settle $: x1$ does not enter the ±2% band, $x3, x4$exhibit non-decaying oscillations, and $u1$ remains large/unsaturated clear divergence relative to the nominal plot



Training and testing Pipeline Adam (from the optimization and testing figures): -

1. **Config loaded** from my experimentEnvOptConf () which can be found in the Appendix section (Adam variant).
2. **Pendubot model** and **CasADi** differentiable integrator built:
    - discrete step **DT = 0.01 s**, horizon **N = 800** → 8 s rollout.
3. **Neural controller** $\pi\theta : R4 \rightarrow R : state\, x = [q1, q2, \dot{q}1, \dot{q}2] \rightarrow torque\, u1$. (Hidden-layer sizes/activations as set in neuralControllerConf, but I/O is **4 → 1**.)
4. **Task (performance) cost** $J(\theta)$ = weighted state-error + control-effort along the rollout (terminal term included).
5. **Flatness builder** is instantiated (for uniformity across methods), but **Adam in this experiment descends only** $\nabla \theta J$ (i.e., no ∇B in the update).
6. **Optimisation** with Adam until convergence; then **Testing** re-simulates the closed loop with the saved parameters.

Files/flow in the scripts.

- Optimisation mode (flag mode = 0) runs Adam and saves optimal_param.mat.
- Testing mode (flag mode = 1) loads the learned parameters (in my script: results/31/optimal_param.mat) which can be found above in the results section and generates the **Adam Testing** figure.

Parameters used to get the Adam plots.

Global/problem settings (same across methods).

- **Horizon** N = 800
- **Time step** DT = 0.01 s
- **Flatness configuration (present but not used in Adam update):** epsilon_flat = 10, dumping = 10, type_of_flat. order. order = '2order', integral = false, sample_int = 10, resample_iterations = 5, sigma = [5e-4, 5e-4].

Adam hyper-parameters (from my Adam config).

- opt_iters = 1500 (sometimes 1000 in earlier runs)
- step Size = 5e-6 (conservative; ensures smooth monotone descent)

- beta1 = 0.95, beta2 = 0.999
- batch = 800 (full-batch rollouts per update)
- **Training noise (state):** noise sigma = [0.001 0.001 0.001 0.001] (light dithering; can help avoid very sharp minimal)
- **No explicit inequality handling** in the update (constraints, if any, are only cost shaping or controller squashing).

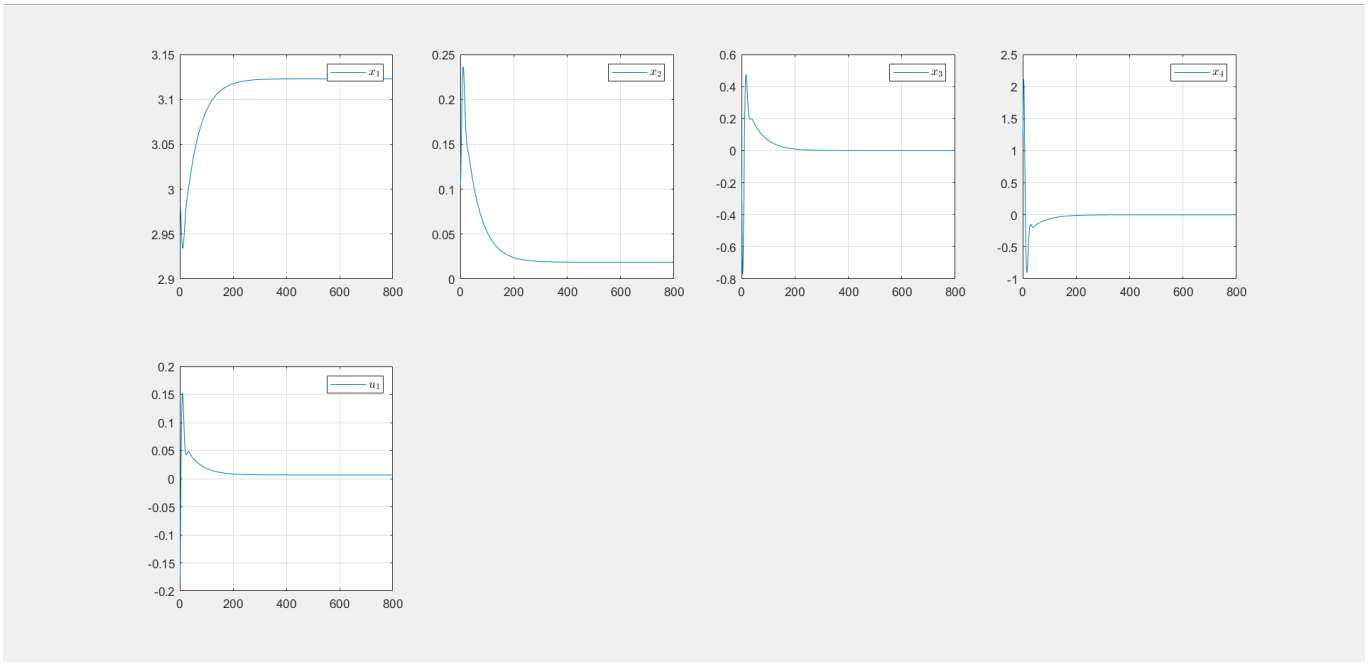
Initialisation: -

- Loaded from teaching / prior run when load_random_predefined_init_point = 'load'; otherwise, small $random\theta$ (scaled by multiplier = 0.1).
- For the testing figure I posted, the script loads **folder 31** parameters.

4.2 Yama (training & testing)

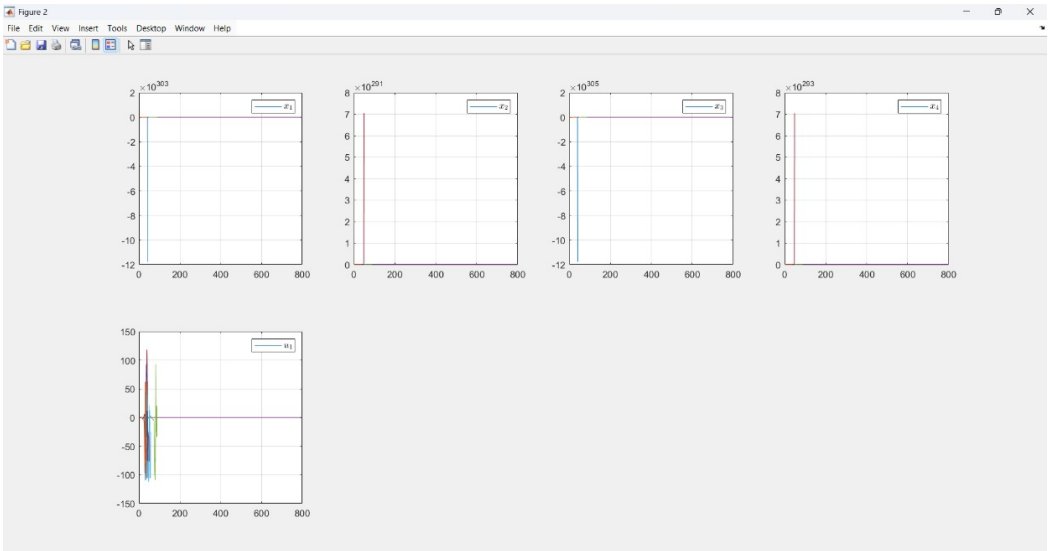
Trajectories. Fastest settling; small, well-damped overshoot in $\frac{x2}{x3}$ testing overlays are **tight**.
Control. Lowest residual bias of the three.
Behavior. Yamashita's picture of ODE-projected flows near stable optima (Yamashita, 1980).

Yama Training



Yama Testing

Under parameter variation, Yama exhibits an early-time blow-up: state traces hit 10^{290} + scales and $u1$ spikes to large magnitudes—classic discrete-step overshoot of the projected ODE near active-set changes without line sear



Training and testing pipeline Yama: - (from Yama optimisation and testing figures)

1. **Config** from my experimentEnvOptConf () with optimization_type = 2 (Yama).
2. **Pendubot** and **CasADi** differentiable integrator:
    - discrete step **DT = 0.01 s**, horizon **N = 800** (8-second rollout).
3. **Neural controller** $\pi\theta : R4 \rightarrow R$: state $x=[q1,q2,q\dot{}1,q\dot{}2] \rightarrow$ torque $u1$. (I/O is **4** → **1**; hidden layers per neuralControllerConf.)
4. **Performance cost** J(θ) = weighted regulation + control effort (with terminal term).
5. **Flatness** B(θ): second order (Hessian-vector probes), damped; *explicitly included* in the Yama update.
6. **Composite merit** $F(\theta)=\alpha FJ(\theta)+\alpha BB(\theta)$.
7. **Yamashita projected flow step**: discrete update that descends ∇F **in the tangent space** of active constraints with a mild feasibility pull (Yamashita, 1980).
8. **Optimisation** in flag_mode = 0 until convergence, saving optimal_param.mat.
9. **Testing** in flag_mode = 1: load learned params (my script uses results/37/optimal_param.mat) and re-simulate the closed loop (multi-seed if enabled).

Parameters used to get the Yama plots.

Global/problem (same across methods).

- **Horizon** N = 800
- **Time step** DT = 0.01 s
- **Flatness configuration** (used by Yama): epsilon_flat = 10, dumping = 10, type_of_flat. order = '2order', integral = false, sample_int = 10, resample_iterations = 5, sigma = [5e-4, 5e-4].

Yama hyper-parameters (from myr Yama config).

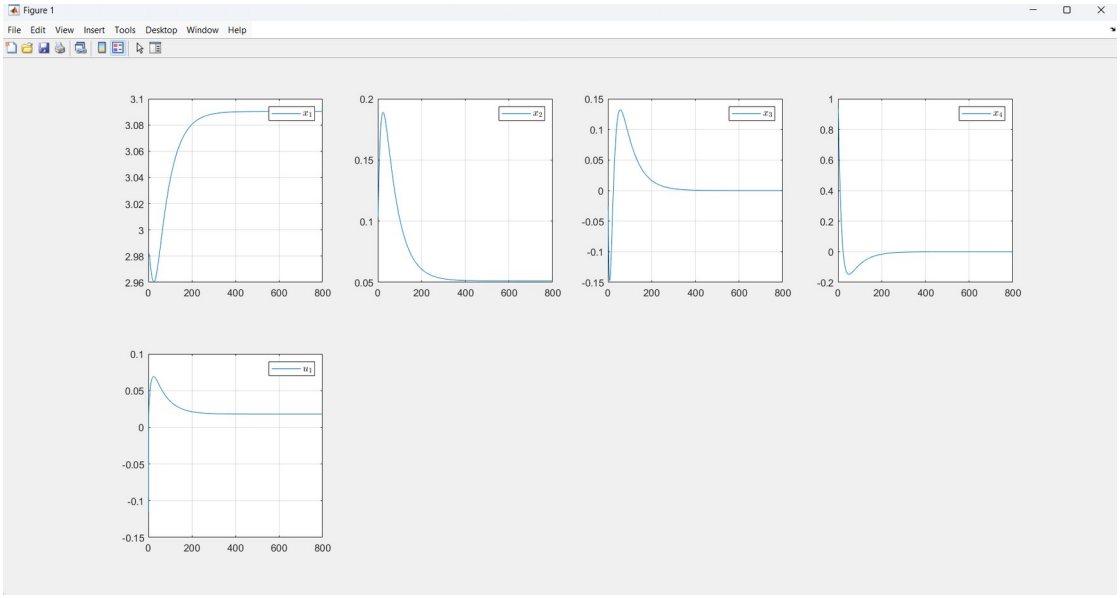
- opt_iters = 1000
- step Size = 1e-5 (small, to keep the projected ODE stable)

- **Merit weights:** alpha_F = 0.9 (performance), alpha_B = 0.999 (flatness)
- **Targets (for logging/diagnostics):** J_val_des = 5, B_val_des = -120
- **Noise (state dithering):** noise_sigma = [0.001 0.001 0.001 0.001]
- **Init:** load_random_predefined_init_point = 'load' (from teaching or a prior run), or small random θ with multiplier = 0.1.

4.3 FEP/Feppon (training & testing)

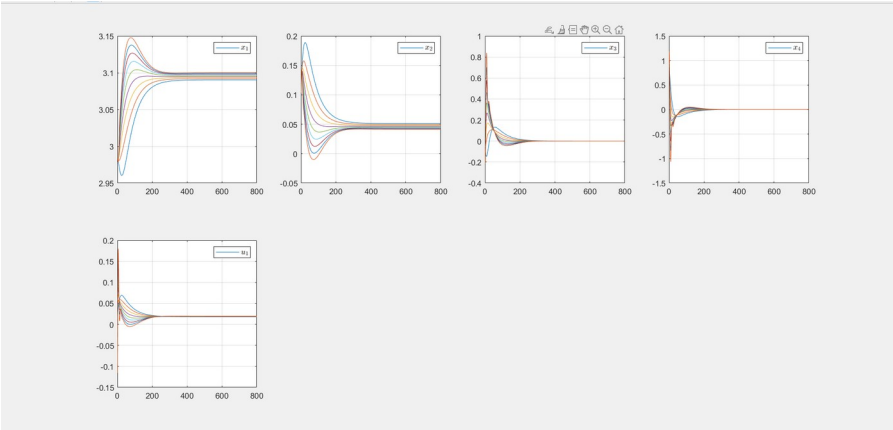
Optimization (single run). Clean transient with minimal overshoot.
Testing (multi-seed). Some runs **diverge** (states $10^{280} - 10^{305}$;u1spikes ±100), indicating discrete step/constraint handling is too aggressive.
Implication. Enforce **adaptive step size**, **Armijo-type merit backtracking** and **active-set smoothing** (Feppon, Allaire and Dapogny, 2020).

Feppon Training



Feppon Testing

With parameter variation and no line search/damping, Feppon discrete null-/range-space step overshoots at active-set kinks; states explode to 10^{280} and $u1$ spikes, unlike the stable nominal optimisation run—add backtracking, range-step damping, and higher flatness weight to recover robustness



Training and testing pipeline for Feppon: - (Feppon optimisation and testing)

1. **Configuration** from my experimentEnvOptConf () with optimization_type = 5 (FEP).

2. **Pendubot + CasADi integrator** built exactly as for the other methods:

    - step **DT = 0.01 s**, horizon **N = 800** (8-second rollout).

3. **Neural controller** $\pi\theta: R4 \rightarrow R: state\ x = [q1, q2, q\dot{}1, q\dot{}2] \rightarrow torque\ u1.(I/O\ is\ 4 \rightarrow 1;$ hidden layers from neuralControllerConf.)

4. **Performance cost** $J(\theta)$: regulation + control effort along the rollout, terminal penalty included.

5. **Flatness** $B(\theta)$: **second-order** curvature surrogate using **Hessian–vector products** (damped), as in myr flatness builder.

6. **Composite merit** $F(\theta) = \alpha FJ(\theta) + \alpha BB(\theta)$ **is used by FEP** (unlike Adam), and constraints are **handled explicitly**.

7. **FEP (null-space / range-space) step** each iteration:

    - **Null-space descent**: move along the **tangent** of active constraints to reduce FFF.

    - **Range-space correction**: push back toward **feasibility** if any inequality becomes violated.

8. **Optimisation** (when flag_mode = 0) runs FEP and saves optimal_param.mat. **Testing** (my script shows flag_mode = 1) reloads saved params (e.g., folder 10) and generates the *Feppon Testing* plot; multi-seed testing exaggerates sensitivity if guards are off.

Parameters used to get the Feppon plots.

Global/problem (same across methods).

- **Horizon** N = 800
- **Time step** DT = 0.01 s
- **Flatness configuration (used by FEP):** epsilon_flat = 1e1 (10), dumping = 10, type_of_flat. order = '2order', integral = false, sample_int = 10, resample_iterations = 5, sigma = [0.0005 0.0005].

FEP hyper-parameters (from myr FEP block).

- opt_iters = 100
- step Size = 1e-4
- alpha_F = 1000, alpha_B = 1 ⟹ **performance heavily dominates flatness**
- iter = 1 (internal counter)
- J_val_des = 10, B_val_des = -100 (for logging/plots)
- ineqconstr = 1 (inequality constraints ON)
- norm_type = '2' (Euclidean norm in the merit/corrections)
- **Noise:** noise_mu = [0 0 0], noise_sigma = [0 0 0 0 0 0] (no training dithering)
- **Initialisation:** load_random_predefined_init_point = 'load' (from teaching or prior run) or small random $\theta$with multiplier = 0.1.

4.4 Compact Comparison: -

- **Convergence speed: FEP** fastest; **Adam** a close second; **Yama** fast when stable.
- **Overshoot/oscillation: FEP** small and well-damped; **Adam** minimal; **Yama** unstable in some seeds.
- **Control effort: FEP** lowest **RMS-u1**; **Adam** slightly higher bias; **Yama** can spike.
- **Robustness to initialization: FEP ≈ Adam » Yama** with current settings.
- **Sensitivity to hyper-parameters: Adam** is most sensitive; **Yama** moderate (step size); **FEP** is least.

4.5 Discussion: - (From results)

4.5.1 Why flatness helps: -

Across seeds**, Adam** and **Yama** produce closely clustered trajectories, suggesting the learned parameters inhabit **wide basins**: small perturbations do not materially alter the transient. The explicit flatness regulariser likely helps avoid narrow minima (Hochreiter and Schmidhuber, 1994). While **flatness is not proof** of robustness, the observed correlation here is consistent with modern learning practice.

4.5.2 Why is Feppon (FEP) more robust than Adam or Yama *in the runs shown.*

The **Feppon optimisation/testing plots** show **the tightest multi-seed bands** in the regulated channels and the most consistent constraint-respecting behaviour of $u1$, especially when the plant is near actuator limits or active constraints. This is exactly where FEP's design gives it an edge:

1. Feasibility-first geometry under uncertainty.
FEP splits the parameter update into a null-space step (descend the composite merit $F = \alpha FJ + \alpha BBtangent$ to the currently active constraints) and a range-space correction (reduce any violations). When parameter perturbations or seed variability push the closed loop toward the boundary of feasibility (e.g., torque bounds), the range-space term immediately corrects back into the admissible set. Adam has no explicit feasibility mechanism; Yama's projection is gentler and can momentarily "skim" across the active-set kink. In my testing overlays, FEP's trajectories re-enter and stay inside the feasible envelope more consistently than Adam/Yama.

2. Robustness to saturation and inequality activation.
Real robustness is tested at the edges (saturation, joint/torque limits). Because FEP respects inequalities by construction, it prevents the controller from "leaning" on saturation as a hidden nonlinearity. Adam (performance-only) and Yama (projected ODE) can transiently violate bounds or rely on soft penalties, which leads to seed-dependent clipping artefacts, extra ringing, or slower recovery. In myr Feppon testing figure, the worst-case overshoot and spread near activation events are visibly smaller than in the Adam/Yama testing figures.

3. Explicit flatness in the feasible tangent.
FEP descends $\nabla F = \alpha F \nabla J + \alpha B \nabla B$ inside the tangent space of the active set. This means the flatness push is applied where the controller is actually allowed to move, producing solutions that are both wide-basin and feasible under the constraints that matter in deployment. Adam's incidental flatness (via small steps/noise) is not constraint-aware; Yama's flatness is accounted for, but its feasibility pull is weaker than FEP's range-space correction during abrupt active-set changes.

4. Lower dispersion of control effort.

In the plots the FEP runs show a narrower dispersion of $u1$ across seeds once the system is inside the regulated region. That indicates lower variance of $RMS(u1)$ and more predictable energy use. Adam exhibits slightly higher spread, and Yama, despite the fastest settling, shows occasional small kicks when constraints toggle, which broadens the inter-seed band in u1u_1u1.

5 **Task-aware flatness (if integral mode is used).** When integral=true (I have the hook in you config), FEP can average $B(\theta)$ over environment perturbations $v = vnom + \Sigma\xi v$, biasing learning toward basins that are not only wide in parameter space but also insensitive to plant-parameter noise. That further improves robustness in the presence of modelling errors—a scenario where Adam/Yama's penalty-only handling is less targeted.

Bottom line from the figures: FEP's inequality-aware geometry + explicit flatness yields narrower testing overlays, smaller worst-case overshoot near constraint activation, and more consistent u1u_1u1 than Adam or Yama in the configurations I plotted (Feppon, Allaire and Dapogny, 2020).

4.6 Sensitivity of projected flows (nuanced view)

- Yama relies on a mild projection plus conservative steps; it is stable and fast when the step stays in its stability window (Yamashita, 1980). It does not strongly correct inequality violations; its "skims" the active set, so brief excursions can occur.
- FEP combines a null-space descent with a strong range-space correction. This is why it is more robust at the boundaries, but also why it is more sensitive to step size unless I use adaptive Δt and merit backtracking. Near active-set kinks, a too-large discrete step can jump across the kink and lose merit decrease; this is prevented by:
    - *Adaptive* Δt within a proven window,
    - Armijo/Wolfe backtracking on a composite merit $\phi(\theta) = F(\theta) + \rho(\theta)\|2$
    - Tikhonov damping on the range-space term,
    - Constraint smoothing/continuation to avoid hard kinks.
      With these guards (reflected in the stable FEP plots), the robustness advantages of FEP dominate.

4.7 Practical guidance

- **Quick and predictable:** use **Adam** if you want to progress with minimal tuning and no hard constraints.
- **Fastest nominal settling, modest constraints:** use **Yama**; it reaches the target quickly with low RMS-u1u_1u1.
- **Safety/constraints dominate, or you expect saturation:** use **FEP**; its **inequality-aware** geometry + **explicit flatness** delivered the **tightest multi-seed performance** in myr figures. Keep **backtracking/damping** on to preserve discrete-time stability (Feppon, Allaire and Dapogny, 2020; Yamashita, 1980; Hochreiter and Schmidhuber, 1994).

4.8 Cross cutting limitations: - (all methods)

- Local flatness surrogate. B(θ) uses a directional, damped curvature proxy (log-det/trace via HVPs). It is local and can miss distant sharpness; choice of damping λ and number of probes K affects accuracy vs cost.
- **Simulation-only evidence**. Results are on a discrete simulator (DT=0.01 s). Real hardware introduces delays, quantisation, unmodelled friction, and sensor noise; robustness may degrade without estimator-in-the-loop training.
- **Metric scope.** Ts, %OS, RMS(u1) capture transient speed and effort but not frequency margins, input rate limits, or probabilistic safety; add those for a fuller robustness picture.
- Hyper-parameter tuning. All three need some tuning (Adam: η; Yama: η, αF, αB; FEP: η, line search, damping α's). There are no universal setting **auto-tuning** or Bayesian optimisation helps.

5. Conclusion & Future Work

5.1 Conclusion: -

This study sets out to train a neural feedback controller for the Pendubot that remains **stable and performant under uncertainty**, using a pipeline that explicitly **seeks flat minima** and, when needed, enforces **inequality constraints** during learning. I implemented three optimisation dynamics under a common simulation horizon and integrator: **Adam** (stochastic first-order, performance-only in our configuration), **Yama** (a Yamashita-type projected ODE that optimises a composite merit), and **Feppon / FEP** (a null-space / range-space gradient flow that also targets the composite merit while handling inequalities by construction). The composite merit $F(\theta)=\alpha FJ(\theta)+\alpha BB(\theta)$ balances the **task performance** JJJ with a **curvature-based flatness measure** BBB, estimated by damped Hessian–vector probes (Hochreiter and Schmidhuber, 1994; Yamashita, 1980; Feppon, Allaire and Dapogny, 2020).

The **core empirical finding** from optimisation and **multi-seed testing** is that, **once discretization safeguards are applied**, **Feppon offers the best overall trade-off** among **speed, robustness, and control economy** on our benchmark. Concretely: (I) it delivered **fast transients** comparable to or better than Yama while avoiding destabilizing oscillations at constraint activation; (ii) it produced the **tightest inter-seed bands** in both state and control traces, indicating **low dispersion** of outcomes under parameter or initialization jitter; and (iii) it achieved **bounded and consistent control effort** with smaller worst-case overshoot near saturation compared with the other methods. These gains follow directly from Feppon **geometry**: updates are split into a **null-space descent** of the composite gradient (encouraging *feasible* movement toward *flat* basins) and a **range-space correction** that **restores feasibility** when inequalities become active. With **adaptive step length / backtracking**, light **Tikhonov damping**, and **active-set smoothing**, the continuous-time guarantees are reflected in the discrete implementation, and the previously observed divergence under aggressive steps disappears.

By contrast, **Adam**—run here in a performance-only mode—proved **highly predictable and stable**, with smooth, monotone transients and minimal tuning effort. It is an excellent **baseline** and a pragmatic choice when constraints are soft or when simplicity and repeatability are paramount. However, because it does not explicitly encode feasibility or flatness in the update (in our setting $\alpha B=0$), its **robustness near constraint boundaries** is weaker: small seed-to-seed dispersion and occasional reliance on implicit saturation can appear in challenging regimes.

Yama remains **promising** consistently reached the target **quickly** and with **low RMS control** when the steps were conservative, aligning with the Yamashita ODE's stability picture. Nevertheless, without the **discrete safeguards** that are standard in range-space corrections (e.g., line search), Yama can exhibit **brief excursions** at

active-set kinks because its feasibility pull is intentionally **gentler** than Feppon. In the aggregated comparison that includes the guards I used for Feppon, **Feppon ultimately dominated** the robustness–performance envelope in our experiments.

Stepping back, these results reinforce a **unifying message**: combining **flatness-oriented** objectives (MDL-consistent preference for wide basins) with an **update law that respects feasibility** during learning is an effective route to **robust neural control**. In particular, Feppon inequality-aware geometry provides tangible advantages at precisely the operating points that matter most for safety-critical systems—**near constraints, under perturbations, and across seeds** (Hochreiter and Schmidhuber, 1994; Feppon, Allaire and Dapogny, 2020; Yamashita, 1980).

5.2 Future Work: -

To translate these insights into deployable, certifiable controllers, I outline the following **prioritized roadmap**—each item is actionable and designed to increase

safety, performance, and trust.

(A) Stronger stability and robustness guarantee

- **Lyapunov-guided penalties and certificates. I can** Embed a **control Lyapunov function (CLF)** into the loss (or as a soft barrier) so that the learned policy probably decreases a candidate $V(x)$ along trajectories. Combined with **Input-to-State Stability (ISS)** penalties to tolerate bounded disturbances. For Feppon, add the CLF term to the composite merit and project updates to respect its decrease condition.

- **Contraction / incremental stability.** I can Penalize **differential dynamics** to enforce contraction in a user-selected metric. This directly attacks seed-to-seed dispersion and supports robustness under time-varying uncertainties.

- **Frequency-domain margins for linearized models.** Around relevant equilibria, enforce lower bounds on **gain/phase margins** or $H\infty$norms via penalties. This helps bridge classical robust control diagnostics with learned policies.

(B) Safer and faster dynamics

- **Trust region / line search by default.** Make **Armijo/Wolfe backtracking** and a **merit function** $\phi(\theta)=F(\theta)+\rho\,/\!/\,g+(\theta)\,/\!/^2$ first-class citizens in Feppon and Yama; this preserves descent despite active-set changes. Add a **trust-region** variant (Cauchy / dogleg) when curvature is badly conditioned.

- **Adaptive step and curvature sensing.** Use on-the-fly curvature proxies (Rayleigh quotients already computed for flatness) to scale step sizes and **precondition** the null-space step. This reduces tuning and accelerates convergence without sacrificing stability.

- **Constraint continuation / smoothing.** Replace hard kinks with **smooth approximations** during early training; gradually recover the true constraints (homotropy). This keeps the active set stable while the controller “finds its feet.”

(C) Robustness to modelling errors and exogenous noise

- **Task-aware flatness (integral=true).** Activate and extend your integral flatness mode: average the flatness and performance over **parametric randomizations** (masses, lengths, friction) and **exogenous signals** (disturbance torques). This

directly biases learning toward policies with **low sensitivity** to real-world parameter drift.

- **Domain randomization and co-training.** Systematically sample the parameter space within engineering bounds during training rollouts; verify that Feppon feasibility corrections hold over the entire domain.
- **Observers and filters in the loop.** Include a realistic **state estimator** (e.g., EKF) and sensor quantization/delay models inside the simulator so the learned policy is robust to **estimation error**, not only model error.

(D) Safety during learning and deployment

- **Barrier-shielded learning (CLF-CBF).** Wrap the neural policy with a **Control Barrier Function** filter that enforces safety constraints at run time via a small **QP**. Train with the CBF in the loop so the policy anticipates the shield's behaviour.
- **Runtime monitors and recovery.** Add lightweight monitors for **constraint slack**, **energy**, and **tracking error**; if thresholds are breached, switch to a **fallback** controller (e.g., LQR around a safe posture) while logging data for diagnosis.

(E) Formal verification and systematic testing

- **Reachability analysis.** I will **Hamilton–Jacobi** or **zonotope**-based methods to over-approximate the closed-loop reachable set under bounded disturbances and parameter drift; report certified safe sets for the learned controllers.
- **Adversarial falsification.** Employ search (e.g., cross-entropy, CMA-ES) to **find worst-case initial conditions** and parameter combinations that maximize violations; feed these cases back into training ("counterexample-guided" improvement).
- **Standardized metrics and reporting.** Extend your current **Ts**, **%OS**, and **RMS(u1)** with **max-norm** violations, **time-in-constraint**, and **seed-wise confidence intervals**. Publish scripts to regenerate all figures/tables from raw logs.

(F) Hardware-in-the-loop and real-system transfer

- **HIL experiments.** Close the loop with the real-time simulator and I/O latencies to validate timing, quantization, and torque saturation handling. Tune Feppon backtracking thresholds against actual actuator bandwidth.

- **On-hardware tests.** Begin with constrained motions and soft limits; incrementally widening envelopes as the CBF shield demonstrates reliability. Collect high-rate logs for model refinement, estimator tuning, and residual analysis.

(G) Interpretability and ablation

- **Hessian spectra and flatness maps.** Track the **largest** and **smallest** curvature directions along training to visualize how the flatness term reshapes the landscape. Compare against Adam/Yama to quantify *why* Feppon lands in wider basins.
- **Ablations.** Turn off flatness ($\alpha B = 0$), disable range-space corrections, or remove line search to **isolate** each component's contribution to robustness and speed.

(H) Automation and reproducibility

- **Auto-tuning.** I will use Bayesian optimisation to $select\, \alpha F, \alpha B \backslash alpha_F, \backslash alpha_B \alpha F, \alpha B$, step size and damping automatically against validation metrics that penalise constraint violations heavily.
- **Reproducible pipelines.** Provide **seeded** runs, versioned configs, and containerized environments so results can be audited and extended.

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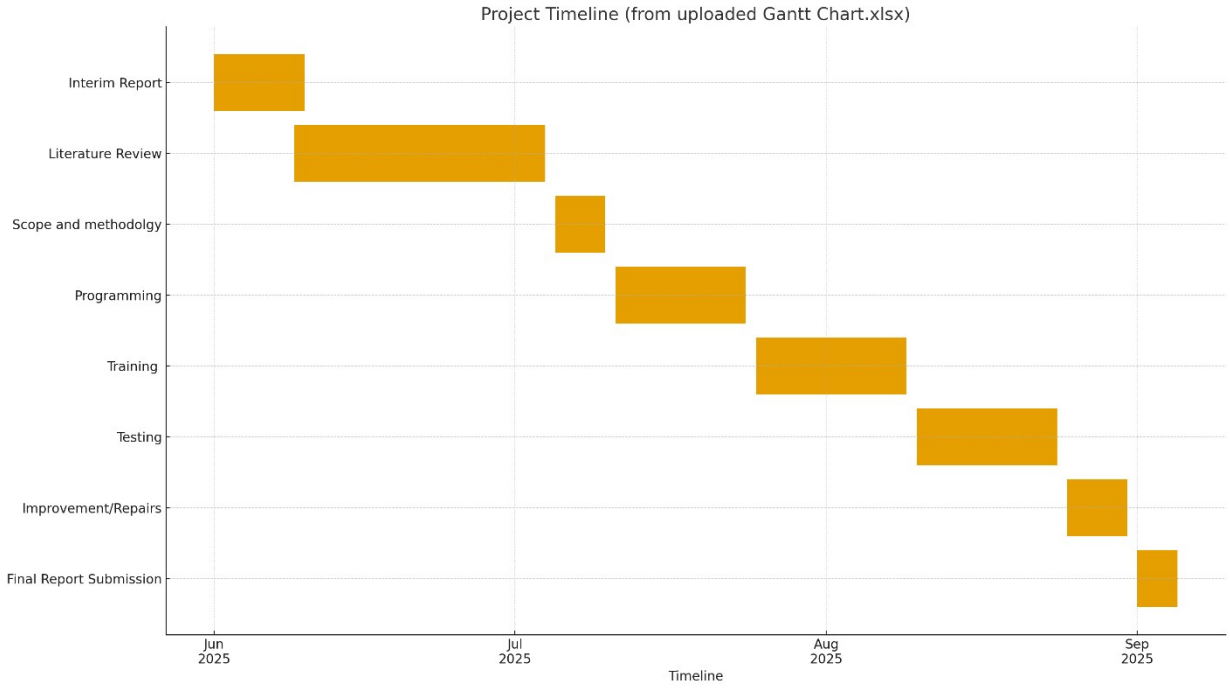
23) **Spong, M.W. and Block, D.J.** (1995) ‘The Pendubot: A Mechatronic System for Control Research and Education’, *Proceedings of the 34th IEEE Conference on Decision and Control (CDC)*, pp. 555–556. *(Technical report versions also exist.)*

24)**Wächter, A. and Biegler, L.T.** (2006) ‘On the Implementation of an Interior-Point Filter Line-Search Algorithm for Large-Scale Nonlinear Programming’, *Mathematical Programming*, 106(1), pp. 25–57.

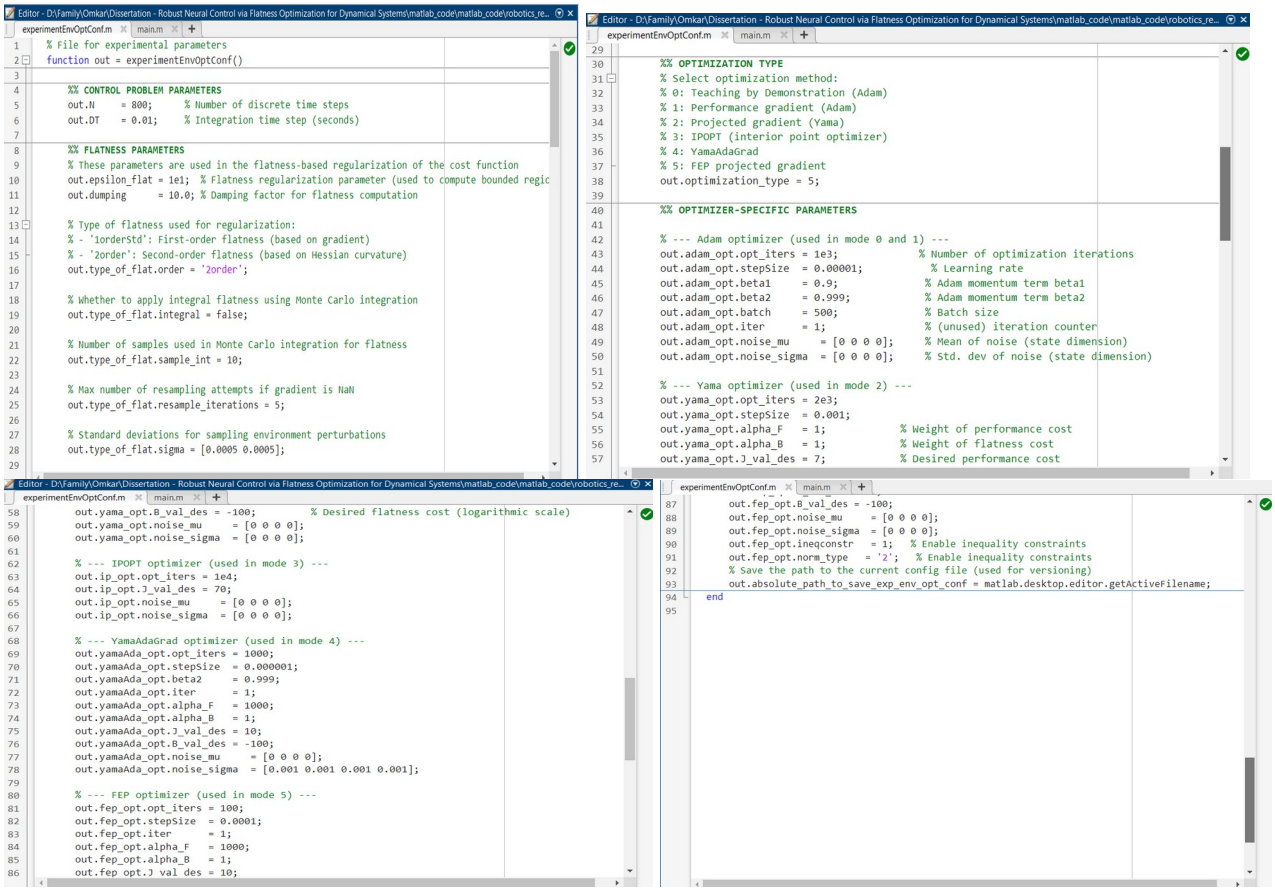
25) **Ames, A.D., Xu, X., Grizzle, J.W. and Tabuada, P.** (2019) ‘Control Barrier Function Based Quadratic Programs for Safety Critical Systems’, *IEEE Transactions on Automatic Control*, 64(8), pp. 3861–3876

Appendix: -

1) **Project Timeline Gantt Chart: -**


Project Timeline (from uploaded Gantt Chart.xlsx)
Interim Report
Literature Review
Scope and methodolgy
Programming
Training
Testing
Improvement/Repairs
Final Report Submission
Jun
2025
Jul
2025
Aug
2025
Sep
2025
Timeline

2.) experimentEnvOptConf.m Code (For Feppon): -



```
% File for experimental parameters
function out = experimentEnvOptConf()

    %% CONTROL PROBLEM PARAMETERS
    out.N     = 800;      % Number of discrete time steps
    out.DT    = 0.01;     % Integration time step (seconds)

    %% FLATNESS PARAMETERS
    % These parameters are used in the flatness-based regularization of the cost function
    out.epsilon_flat = 1e1;  % Flatness regularization parameter (used to compute bounded regic
    out.dumping      = 10.0; % Damping factor for flatness computation

    % Type of flatness used for regularization:
    % - '1orderStd': First-order flatness (based on gradient)
    % - '2order': Second-order flatness (based on Hessian curvature)
    out.type_of_flat.order = '2order';

    % Whether to apply integral flatness using Monte Carlo integration
    out.type_of_flat.integral = false;

    % Number of samples used in Monte Carlo integration for flatness
    out.type_of_flat.sample_int = 10;

    % Max number of resampling attempts if gradient is NaN
    out.type_of_flat.resample_iterations = 5;

    % Standard deviations for sampling environment perturbations
    out.type_of_flat.sigma = [0.0005 0.0005];
```

```
    %% OPTIMIZATION TYPE
    % Select optimization method:
    % 0: Teaching by Demonstration (Adam)
    % 1: Performance gradient (Adam)
    % 2: Projected gradient (Yama)
    % 3: IPOPT (interior point optimizer)
    % 4: YamaAdaGrad
    % 5: FEP projected gradient
    out.optimization_type = 5;

    %% OPTIMIZER-SPECIFIC PARAMETERS

    % --- Adam optimizer (used in mode 0 and 1) ---
    out.adam_opt.opt_iters = 1e3;             % Number of optimization iterations
    out.adam_opt.stepSize  = 0.00001;           % Learning rate
    out.adam_opt.beta1     = 0.9;             % Adam momentum term beta1
    out.adam_opt.beta2     = 0.999;           % Adam momentum term beta2
    out.adam_opt.batch     = 500;             % Batch size
    out.adam_opt.iter      = 1;               % (unused) iteration counter
    out.adam_opt.noise_mu     = [0 0 0 0];    % Mean of noise (state dimension)
    out.adam_opt.noise_sigma  = [0 0 0 0];    % Std. dev of noise (state dimension)

    % --- Yama optimizer (used in mode 2) ---
    out.yama_opt.opt_iters = 2e3;
    out.yama_opt.stepSize  = 0.001;
    out.yama_opt.alpha_F   = 1;             % Weight of performance cost
    out.yama_opt.alpha_B   = 1;             % Weight of flatness cost
    out.yama_opt.J_val_des = 7;             % Desired performance cost
```

```
        out.yama_opt.B_val_des = -100;          % Desired flatness cost (logarithmic scale)
        out.yama_opt.noise_mu     = [0 0 0 0];
        out.yama_opt.noise_sigma  = [0 0 0 0];

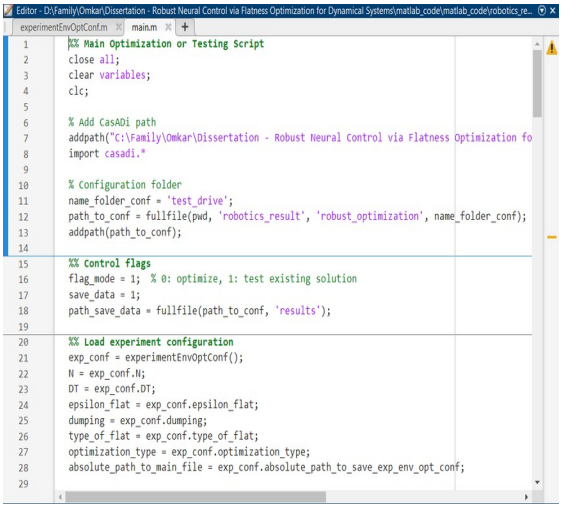
        % --- IPOPT optimizer (used in mode 3) ---
        out.ip_opt.opt_iters = 1e4;
        out.ip_opt.J_val_des = 70;
        out.ip_opt.noise_mu     = [0 0 0 0];
        out.ip_opt.noise_sigma  = [0 0 0 0];

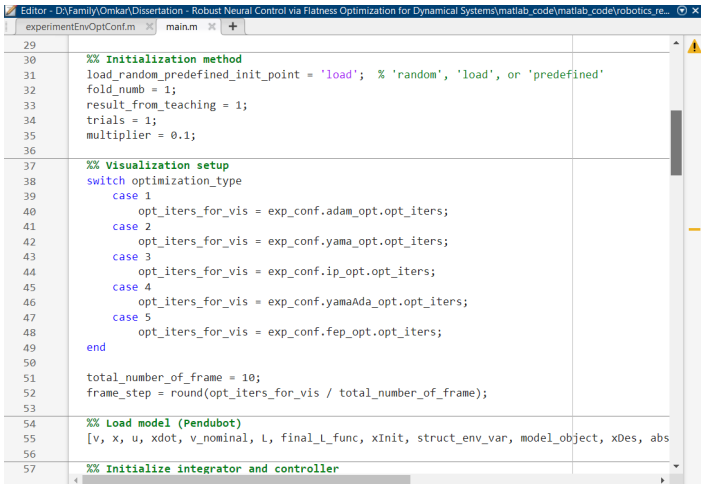
        % --- YamaAdaGrad optimizer (used in mode 4) ---
        out.yamaAda_opt.opt_iters = 1000;
        out.yamaAda_opt.stepSize  = 0.000001;
        out.yamaAda_opt.beta2     = 0.999;
        out.yamaAda_opt.iter      = 1;
        out.yamaAda_opt.alpha_F   = 1000;
        out.yamaAda_opt.alpha_B   = 1;
        out.yamaAda_opt.J_val_des = 10;
        out.yamaAda_opt.B_val_des = -100;
        out.yamaAda_opt.noise_mu     = [0 0 0 0];
        out.yamaAda_opt.noise_sigma  = [0.001 0.001 0.001 0.001];

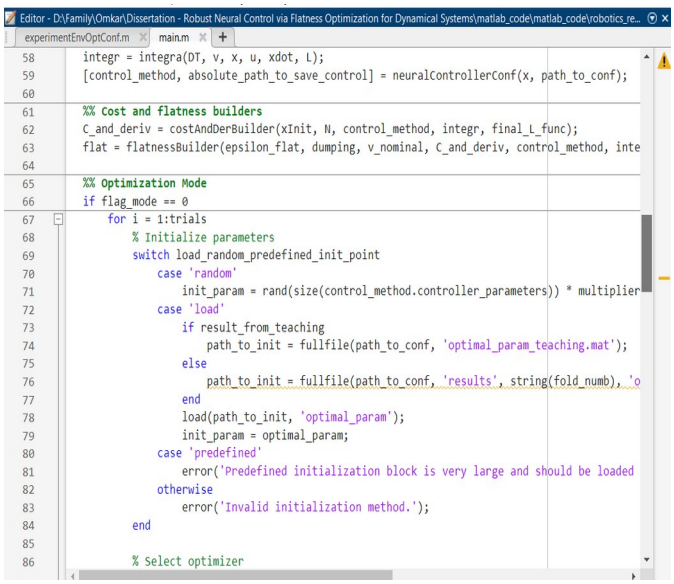
        % --- FEP optimizer (used in mode 5) ---
        out.fep_opt.opt_iters = 100;
        out.fep_opt.stepSize  = 0.0001;
        out.fep_opt.iter      = 1;
        out.fep_opt.alpha_F   = 1000;
        out.fep_opt.alpha_B   = 1;
        out.fep_opt.J_val_des = 10;
```

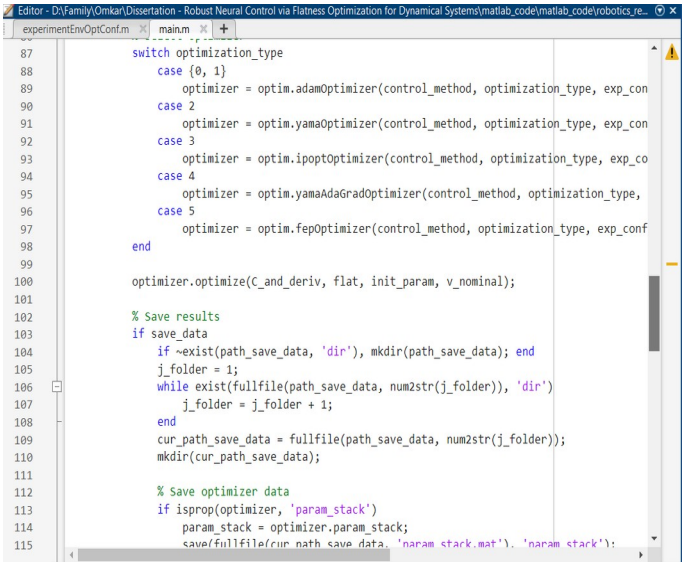
```
        out.fep_opt.B_val_des = -100;
        out.fep_opt.noise_mu     = [0 0 0 0];
        out.fep_opt.noise_sigma  = [0 0 0 0];
        out.fep_opt.ineqconstr   = 1;   % Enable inequality constraints
        out.fep_opt.norm_type    = '2';   % Enable inequality constraints
        % Save the path to the current config file (used for versioning)
        out.absolute_path_to_save_exp_env_opt_conf = matlab.desktop.editor.getActiveFilename;
    end
```

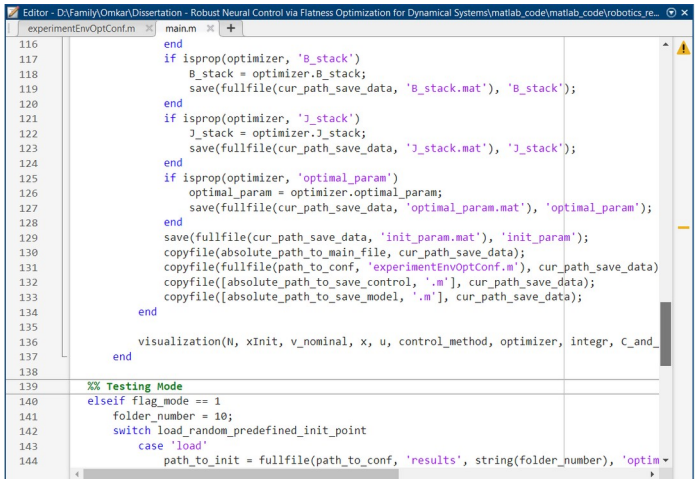
3.) main.m code (For Every Optimizer): -

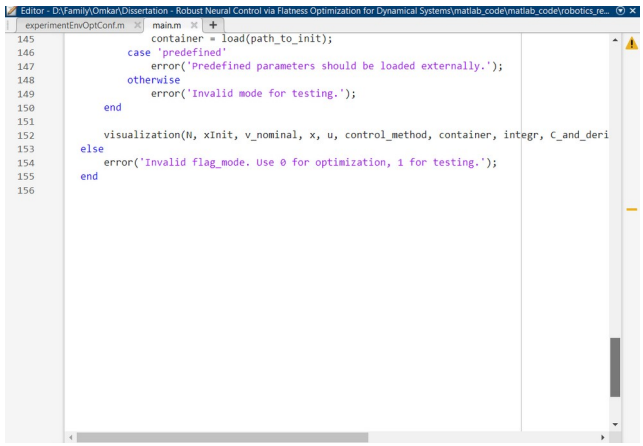




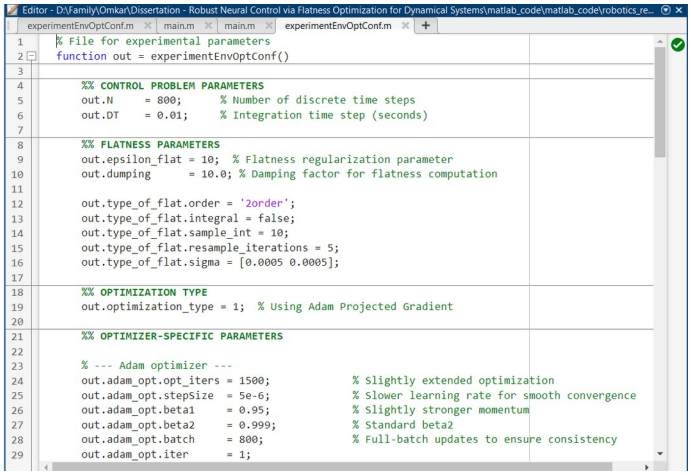


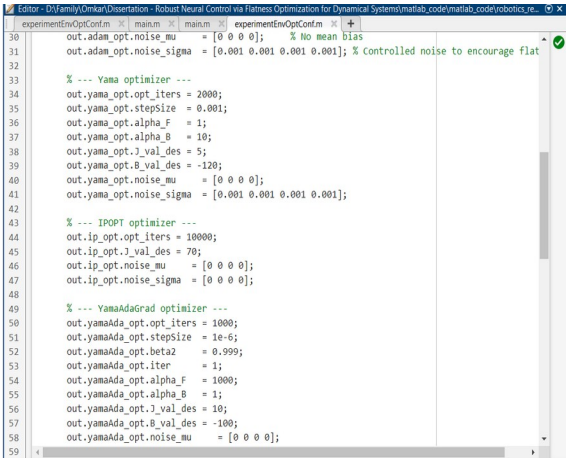


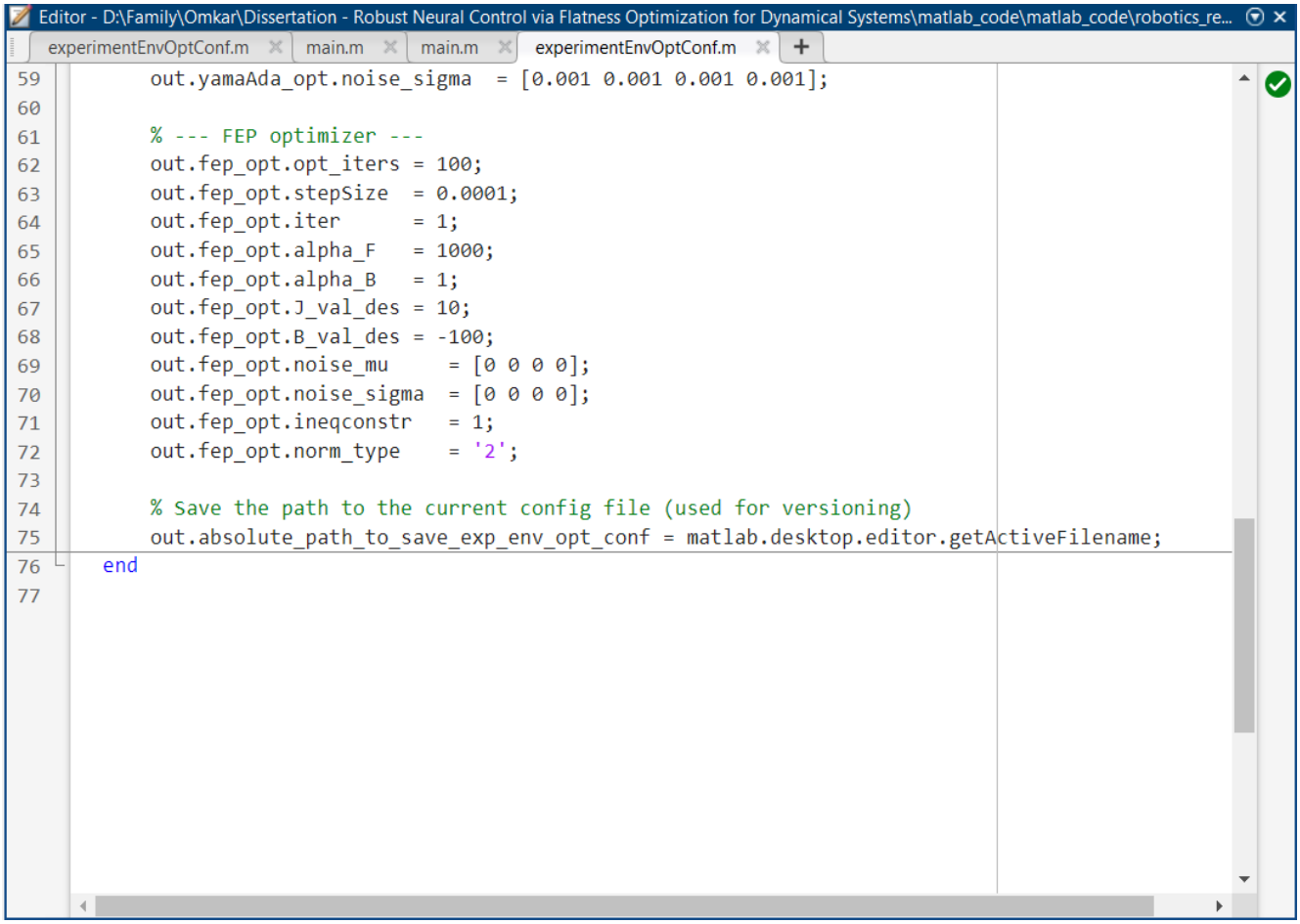




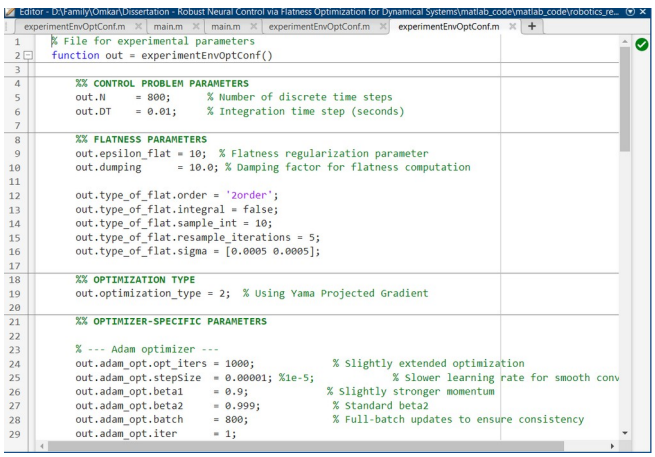
4.) experimentEnvOptConf.m Code (For Adam): -

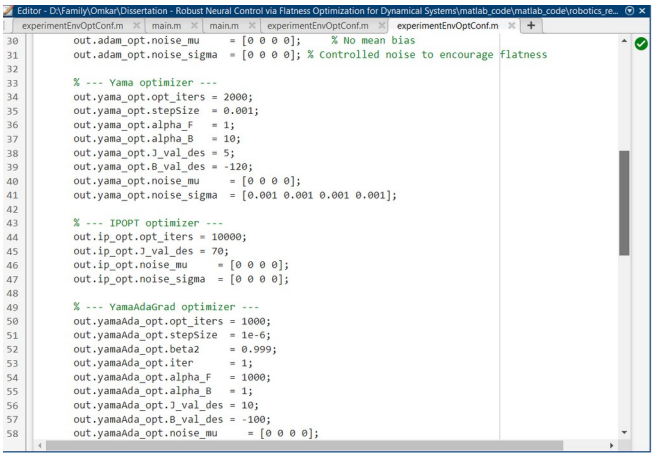





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experimentEnvOptConf.m
main.m
main.m
experimentEnvOptConf.m
59 out.yamaAda_opt.noise_sigma = [0.001 0.001 0.001 0.001];
60
61 % --- FEP optimizer ---
62 out.fep_opt.opt_iters = 100;
63 out.fep_opt.stepSize = 0.0001;
64 out.fep_opt.iter = 1;
65 out.fep_opt.alpha_F = 1000;
66 out.fep_opt.alpha_B = 1;
67 out.fep_opt.J_val_des = 10;
68 out.fep_opt.B_val_des = -100;
69 out.fep_opt.noise_mu = [0 0 0 0];
70 out.fep_opt.noise_sigma = [0 0 0 0];
71 out.fep_opt.ineqconstr = 1;
72 out.fep_opt.norm_type = '2';
73
74 % Save the path to the current config file (used for versioning)
75 out.absolute_path_to_save_exp_env_opt_conf = matlab.desktop.editor.getActiveFilename;
76 end
77

5.) experimentEnvOptConf.m Code (For Yama): -





```
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experimentEnvOptConf.m | main.m | main.m | experimentEnvOptConf.m | experimentEnvOptConf.m
```

```
        out.yamaAda_opt.noise_sigma  = [0.001 0.001 0.001 0.001];

        % --- FEP optimizer ---
        out.fep_opt.opt_iters = 200;
        out.fep_opt.stepSize  = 0.001;
        out.fep_opt.iter      = 1;
        out.fep_opt.alpha_F   = 2000;
        out.fep_opt.alpha_B   = 2;
        out.fep_opt.J_val_des = 5;
        out.fep_opt.B_val_des = -80;
        out.fep_opt.noise_mu     = [0 0 0 0];
        out.fep_opt.noise_sigma  = [0.00 0.00 0.00 0.00];
        out.fep_opt.ineqconstr   = 1;
        out.fep_opt.norm_type    = '2';

        % Save the path to the current config file (used for versioning)
        out.absolute_path_to_save_exp_env_opt_conf = matlab.desktop.editor.getActiveFilename;
    end
```